

Utilization of environmental and epidemiological indicators in the study of malaria dynamics

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetorias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes play a central role in shaping malaria transmission. In the Amazon, processes such as deforestation, agricultural expansion, and infrastructure development alter local ecosystems, affecting vector habitats and increasing human–vector contact. These changes often promote the proliferation of mosquito breeding sites and concentrate human populations in high-risk areas, particularly among non-immune migrants [4, 5]. In addition to land-use change, climatic variables such as temperature and rainfall, as well as disturbances such as forest fires and mining activities, further influence vector abundance, pathogen development, and transmission intensity.

Despite the recognized importance of environmental drivers, integrating these factors into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human–vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we develop and analyze a modified SIR–SEI

model for malaria transmission in the Amazon, incorporating environmental variables such as temperature, rainfall, deforestation, and fire activity. Climatic factors are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Using epidemiological and environmental data from the Trajetorias Project [6], we investigate how these factors influence disease dynamics and assess their relative contributions through analytical and numerical methods.

2. Model Formulation

2.1. Baseline Delayed SIR–SEI Model

We begin with the classical host-vector framework for malaria transmission, based on the model introduced by Parham and Michael (2010) [7], which incorporates temperature- and rainfall-dependent mosquito dynamics. The human population is divided into susceptible, infected, and recovered compartments, denoted by S_H , I_H , and R_H , respectively. The mosquito population is structured into susceptible, exposed, and infectious compartments, denoted by S_M , E_M , and I_M .

The transmission dynamics are governed by the system of delay differential equations:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M(t)}{N} S_H(t) \quad (1a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M(t - \tau_H)}{N} S_H(t - \tau_H) - \gamma I_H(t) \quad (1b)$$

$$\frac{dR_H}{dt} = \gamma I_H(t) \quad (1c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H(t)}{N} + \mu(T) \right) S_M(t) \quad (1d)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \frac{I_H(t)}{N} S_M(t) - \mu(T) E_M(t) \\ & - ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) \end{aligned} \quad (1e)$$

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$$\frac{dI_M}{dt} = ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) - \mu(T) I_M(t) \quad (1f)$$

Here, $a(T)$ denotes the biting rate, b_1 and b_2 are transmission probabilities from humans to mosquitoes and vice versa, $\mu(T)$ is the mosquito mortality rate, and τ_H , τ_M represent the intrinsic and extrinsic incubation periods. The function $b(R, T)$ describes mosquito recruitment, while $\ell(\tau_M)$ represents survival during the extrinsic incubation period.

Although this formulation incorporates biologically relevant delays, numerical simulations revealed that, under the parameterization considered, the model fails to reproduce sustained transmission dynamics. In particular, both the infected human population and the mosquito population tends to extinction, indicating that the structure does not adequately capture endemic persistence.

2.2. ODE approximation and structural limitations

To simplify the analysis and avoid the technical complexity of delay differential equations, an ordinary differential equation (ODE) approximation was initially considered. In this formulation, the extrinsic incubation period in mosquitoes is modeled via a transition rate b_3 , replacing the explicit delay τ_M .

The resulting system is given by:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M}{N} S_H \quad (2a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M}{N} S_H - \gamma I_H \quad (2b)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (2c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H}{N} + \mu \right) S_M \quad (2d)$$

$$\frac{dE_M}{dt} = ab_1 \frac{I_H}{N} S_M - (\mu + b_3 \ell) E_M \quad (2e)$$

$$\frac{dI_M}{dt} = b_3 \ell E_M - \mu I_M \quad (2f)$$

While this formulation restores positivity and simplifies numerical implementation, it exhibits a fundamental limitation: the system tends toward a disease-free equilibrium after a single outbreak, failing to capture the recurrent nature of malaria transmission observed in endemic regions. This behavior highlights the importance of explicitly modeling incubation periods and immunity dynamics.

2.3. SEIRS-SEI model with ODE approximation

To address the limitations of the previous formulations, we introduce an ODE based SEIRS-SEI model that incorporates both incubation delays as done above and temporary immunity in the human population. The human

compartments are extended to include an exposed class E_H , and recovered individuals lose immunity at rate ω , returning to the susceptible class.

To account for incubation dynamics in both hosts and vectors, the transition parameters were defined as temperature-dependent development rates.

For humans, the intrinsic incubation process was modeled through $b_{3H} = \frac{1}{\tau_H}$, where τ_H denotes the intrinsic incubation period.

For mosquitoes, the constant transition parameter $b_3 = 1/\tau_M$ from the original formulation was replaced by a Brière-type function,

$$b_{3M}(T) = \eta T(T - T_{\min})(T_{\max} - T)^{1/2}, \quad (3)$$

which incorporates minimum and maximum thermal thresholds for sporogonic development [42, 43].

The scaling coefficient η was estimated by imposing $b_{3M}(T_{\text{opt}}) = \frac{1}{\tau_M(T_{\text{opt}})}$, yielding

$$\eta = \frac{\frac{1}{\tau_M(T_{\text{opt}})}}{T_{\text{opt}}(T_{\text{opt}} - T_{\min})(T_{\max} - T_{\text{opt}})^{1/2}} \quad (4)$$

Another addition was the inclusion of a humidity-dependent survival component $p(H)$ on to the daily survival rate of the adult mosquitoes, which is modeled as a sigmoid function of the daily minimum relative humidity U . The parameter H_0 (in %) represents the humidity threshold at which the mosquito experiences a 50% reduction in survival relative to its maximum, and k (in $\%^{-1}$) controls the steepness of the transition. The parameter φ (dimensionless) establishes a baseline survival floor, accounting for the persistence of individuals in favorable micro-habitats even under extreme desiccation conditions.

The SEIRS-SEI complete model is then given by:

$$\frac{dS_H}{dt} = -\lambda_H(t) S_H + \omega R_H \quad (5a)$$

$$\frac{dE_H}{dt} = \lambda_H(t) S_H - b_{3H} E_H \quad (5b)$$

$$\frac{dI_H}{dt} = b_{3H} E_H - \gamma I_H \quad (5c)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (5d)$$

$$\frac{dS_M}{dt} = \Lambda(t) - \lambda_M(t) S_M - \mu S_M \quad (5e)$$

$$\frac{dE_M}{dt} = \lambda_M(t) S_M - (\mu + b_{3M} \ell) E_M \quad (5f)$$

$$\frac{dI_M}{dt} = b_{3M} \ell E_M - \mu I_M \quad (5g)$$

The forces of infection are defined as:

$$\lambda_H(t) = a b_2 \frac{I_M(t)}{N(t)} \quad (6a)$$

$$\lambda_M(t) = a b_1 \frac{I_H(t)}{N(t)} \quad (6b)$$

2.4. Recalibration of the temperature-dependent biting rate

In the original formulation [7], the biting rate is expressed as a temperature-dependent function derived from the duration of the gonotrophic cycle. Although this formulation captures the general thermal dependence of mosquito feeding behavior, its parameterization was not developed specifically for Amazonian mosquito populations. With this in mind, we recalibrated the biting-rate function using entomological observations reported by de Barros, Honório and Arruda [46], regarding females of *Anopheles darlingi* collected between August 2003 and July 2004 through intermittent sampling carried out in selected months over the study period.

Their results allowed for estimation of the duration of the gonotrophic cycle through the following formula:

$$P = Q^{1/g} \implies g = \frac{\ln(Q)}{\ln(P)}, \quad (7)$$

where P is the daily survival rate of mosquitoes, Q is the proportion of parous mosquitoes females in the population and g is the duration of the gonotrophic cycle, given in days [47].

Given the lack of more recent data, it was decided to use the obtained estimates of the time period specified above with average temperatures from the period of analysis between 2017 and 2023, given that the biting rate is given as a function of temperature. It was decided to consider $T' = 16^\circ\text{C}$, given that below this temperature there is no development of the vector, as the threshold temperature for the biting rate, and we estimated values of g for the missing months of the analysis via interpolation. With this, given the closed formula for the biting rate, and knowing that it is the inverse of the gonotrophic cycle, we calculated $D_1 = g(T - T')$ for each of the months with available data¹.

Having monthly values of D_1 , we took their average in order to estimate the biting rate and the proportion of female mosquitoes to humans, which is given by

$$m = \frac{HBR}{a(T)}, \quad (8)$$

where HBR is the mean biting density of *An. darlingi*, given in bites/person/night, while $a(T)$ is the biting rate, given in bites/mosquito/night. Based on the data, M_0 , the initial population of mosquitos used in the analysis, was estimated

to be approximately $10N_0$, based on the found value of m for the month of January.

At this point, we also approximated the ratio of initially exposed to infectious mosquitoes to be their values at the steady state:

$$\frac{dI_M}{dt} = b_{3M} \ell E_M - \mu I_M = 0 \implies \frac{E(0)}{I(0)} = \frac{\mu(0,0)}{b_{3M}(0)\ell(0,0)} \approx 3$$

This adjustment preserves the functional structure proposed by Parham and Michael while adapting the parameter values to better reflect the feeding behavior of the primary malaria vector in the Amazon region.

2.5. Density-dependent mosquito recruitment and carrying capacity

Mosquito population dynamics were initially modeled using a linear birth–death process of the form

$$\frac{dM}{dt} = b - \mu M, \quad (9)$$

where b represents the birth rate and μ is the per capita mortality rate. However, this formulation does not account for environmental limitations on population growth.

To incorporate the effect of limited environmental resources, we introduce a density-dependent recruitment mechanism inspired by the classical Verhulst logistic model [40],

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K}\right) - \mu M, \quad (10)$$

where $M(t) = S_M(t) + E_M(t) + I_M(t)$ denotes the total mosquito population, b the per capita recruitment rate, μ the mortality rate, and K the environmental carrying capacity. This formulation captures the reduction in population growth as the mosquito population approaches the environmental limit.

In the present model, this idea is adapted by replacing the constant recruitment term with the density-dependent function

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right), \quad (11)$$

where the environmental dependence is separated into two distinct mechanisms:

1. The climatic modulation of mosquito recruitment through $b(R, T)$, inherited from the original Parham–Michael formulation;
2. The environmental limitation imposed by habitat availability through the carrying capacity $K(t)$.

In this interpretation, $b(R, T)$ governs the rate at which mosquitoes are produced under favorable temperature and rainfall conditions, whereas $K(t)$ determines the maximum

¹ $a(T)$ is defined in Table A.1

sustainable mosquito population supported by the environment. Consequently, temperature and precipitation affect mosquito dynamics both through reproductive success and through the availability of suitable breeding habitats.

The environmental carrying capacity is defined by

$$K(t) = M' \cdot f_T(T(t)) \cdot f_R(R(t)), \quad (12)$$

where M' denotes the baseline mosquito population size. Given that we require $0 \leq 1 - \frac{M(t)}{K(t)} \leq 1$, we would require M' such that $M' \geq \frac{M}{f_T(T)f_R(R)}$.

The environmental response functions are given by

$$f_T(T) = \min\left(1.0, \frac{T(T - T_{\min A.})\sqrt{T_{\max A.} - T}}{T_{\text{opA.}}(T_{\text{opA.}} - T_{\min A.})\sqrt{T_{\max A.} - T_{\text{opA.}}}}\right), \quad (13a)$$

$$f_R(R) = \left(\frac{R}{R_L/2}\right) \exp\left(1 - \frac{R}{R_L/2}\right), \quad (13b)$$

where $f_T(T)$ describes the temperature suitability for mosquito persistence and $f_R(R)$ represents the dual effect of rainfall on habitat availability.

The rainfall response function increases with precipitation under moderate rainfall conditions, reflecting the formation of breeding sites, but decreases under excessive rainfall due to habitat flushing and destruction of immature stages.

Both environmental response functions are normalized so that $0 \leq f_T(T), f_R(R) \leq 1$, ensuring that the carrying capacity remains bounded by the baseline population scale M_0 .

Unlike the classical logistic formulation, where population growth is directly proportional to the current mosquito abundance, the present representation assumes that recruitment is primarily determined by environmental suitability and breeding-site availability. The factor $\left(1 - \frac{M(t)}{K(t)}\right)$ therefore imposes density-dependent regulation by reducing recruitment as the mosquito population approaches the carrying capacity, providing a biologically realistic upper bound on mosquito abundance. The mathematical implications of this formulation are analyzed in Section 5.1.

3. Data Description and Processing

3.1. Epidemiological Data

This study focuses on malaria cases caused by *Plasmodium vivax*, which accounts for the majority of malaria infections in Brazil and is the predominant species in the Amazon region [11, 12, 13].

Epidemiological data were obtained from the SIVEP-Malaria system [18], which provides daily records of notified malaria

cases for all Brazilian municipalities from 2003 to 2023. The analysis was restricted to the period 2017–2023, for which complete epidemiological and environmental data were available.

3.2. Population Estimates

The municipality of Manaus was selected due to its high incidence of *P. vivax* malaria, particularly in rural areas. However, detailed yearly estimates of rural population were not directly available from census data. To address this limitation, population estimates were derived using data from the Trajetorias Project [6], which provides incidence values aggregated over five-year periods.

Incidence is defined as

$$\text{Inc}(d, m, z, t_1, t_2) = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \cdot (t_2 - t_1)} \times 10^5, \quad (14)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ denotes the number of reported cases for disease d in municipality m and zone z over the interval $[t_1, t_2]$, and $\text{Pop}(m, z, \cdot)$ represents the population at the midpoint of the interval.

This relationship was inverted to estimate rural population values from reported incidence and case counts. Using this approach, the rural population of Manaus was estimated to represent less than 1% of the total population, ranging between 0.3% and 0.6% of the total municipal population, while accounting for a disproportionately large share of malaria cases (approximately 30% to 40% over the study periods).

Yearly total population values for Manaus were obtained via interpolation based on official census data from the Brazilian Institute of Geography and Statistics (IBGE) for the years 2000, 2010, and 2022 [15, 16, 17]. Intermediate yearly values were constructed using linear interpolation, while values beyond 2022 were extrapolated based on observed trends.

3.3. Data Consistency Adjustments

During preprocessing, inconsistencies were identified in the Trajetorias dataset. In particular, incidence rates for the period 2015–2019 had been calculated using population estimates corresponding to 2004–2008, resulting in a temporal mismatch. These values were recalculated using population estimates consistent with the appropriate time interval.

Additionally, rural and urban population estimates for intermediate years had been derived using fixed proportions based on 2010 data. To better account for demographic changes, particularly urbanization, these estimates were revised. The rural population for 2006 was approximated as the mean between the reported values for 2000 and 2010, while the 2017 value was obtained through extrapolation based on the observed growth trend. These adjustments

ensured temporal consistency in the population data used throughout the study.

4. Impacts of Environmental Factors

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria transmission in the Amazon region [20]. The removal of forest canopy alters local microclimatic conditions, increasing sunlight exposure and ambient temperature, which in turn favor the development of mosquito larvae and pupae. In particular, the primary malaria vector in the Amazon, *Anopheles darlingi*, is known to thrive in partially deforested environments, where sunlit water bodies promote larval growth and increase the availability of food sources such as algae [21, 22].

Beyond ecological effects, deforestation also influences human exposure to malaria. Activities such as logging, agriculture, and infrastructure development lead to increased human presence in recently altered environments, often in close proximity to mosquito breeding sites. This results in higher human–vector contact rates and facilitates transmission, especially in areas where both vector density and environmental suitability are elevated. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation.

Forest fragmentation has been shown to accelerate mosquito development cycles by increasing temperature and light availability, reducing the time from larval to adult stages and increasing vectorial capacity and survivorship [27, 28]. Similarly, forest fires, often used as a land-clearing mechanism, remove vegetation and modify habitat structure, contributing to long-term ecological changes that can influence mosquito populations and disease transmission dynamics [29, 30]. These processes may also increase the likelihood of subsequent fires by promoting understory drying and fuel accumulation [31].

In this context, deforestation and fire activity are incorporated into the model as external drivers that modify transmission dynamics. Rather than directly altering mosquito physiological parameters, these factors are interpreted as influencing human–vector contact rates and exposure, thereby affecting the effective transmission of the disease.

Deforestation data were obtained from the DETER system, which provides daily records of deforestation alerts by degradation type, reported in km² [34]. The data were aggregated to produce a complete daily time series of deforested area. Forest fire data were obtained from the Brazilian National Institute for Space Research (INPE), which provides daily records of detected fire hotspots [35]. Due to the absence of area-based measurements at this

temporal resolution, the number of detected hotspots was used as a proxy for fire activity.

5. Mathematical Analysis of the Model

5.1. Mosquito population dynamics

In this section, we analyze the qualitative impact of the density-dependent recruitment term on mosquito population dynamics. We compare the proposed formulation with the classical Verhulst model and show that the modified structure prevents extinction under suitable environmental conditions.

Proposition 1 (Environmental persistence of the mosquito population). *Assume that $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$ for all $t \geq 0$. Then, the mosquito population $M(t)$ governed by*

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right) - \mu M(t) \quad (15)$$

cannot converge to zero as $t \rightarrow \infty$. In particular, $M = 0$ is not an equilibrium and cannot be asymptotically stable.

Proof. We begin by recalling the classical logistic model with mortality:

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K} \right) - \mu M. \quad (16)$$

In this formulation, the right-hand side vanishes at $M = 0$, so that $M = 0$ is always an equilibrium. Depending on the parameters, this equilibrium may be stable, allowing extinction of the population.

In contrast, in the present model the recruitment term is given by

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right), \quad (17)$$

and the population dynamics satisfy

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right) - \mu M(t). \quad (18)$$

Evaluating the right-hand side at $M = 0$, we obtain

$$\left. \frac{dM}{dt} \right|_{M=0} = b(R, T) K(t). \quad (19)$$

By assumption, $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$, hence

$$\left. \frac{dM}{dt} \right|_{M=0} \geq b_0 K_0 > 0. \quad (20)$$

Therefore, $M = 0$ is not an equilibrium of the system.

Moreover, for $M(t)$ sufficiently small, we have

$$\frac{dM}{dt} \geq b_0 K_0 - \mu M(t). \quad (21)$$

Choosing $\varepsilon > 0$ such that $\varepsilon < \frac{b_0 K_0}{\mu}$, it follows that

$$\frac{dM}{dt} > 0 \quad \text{whenever } 0 < M(t) < \varepsilon.$$

Thus, any trajectory entering a neighborhood of zero is driven away from it.

Consequently, $M(t)$ cannot converge to zero as $t \rightarrow \infty$, and the extinction state is not asymptotically stable. $\square$

5.2. Basic Reproduction Number ($\mathcal{R}_0$)

To derive the $\mathcal{R}_0$ of the model, we define X_s as the set of all disease-free states,

$$X_s = \{x \geq 0 | x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^T$, such that $x_i \geq 0$ represents the number of individuals in each compartment, and each function is assumed to be continuously differentiable at least twice in each variable (C^2) [36].

Rearranging the equations so that the first m contain the infected individuals, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ be the rate of individuals entering compartment i by other means, and $\mathcal{V}_i^-(x)$ be the rate of individuals leaving compartment i . The disease transmission model consists of non-negative initial conditions together with the following system of equations:

$$\dot{x} = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), i = 1, \dots, n,$$

where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. Here, $F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right]$ and $V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right]$, where x_0 is a Disease-Free Equilibrium (DFE), and $1 \leq i, j \leq m$, and $\mathcal{R}_0 = \rho(FV^{-1})$.

The formulation for $\mathcal{R}_0$ in the complete SEIRS/SEI system is given by:

$$\mathcal{R}_0 = \sqrt{\frac{a^2 b_1 b_2 \Lambda b_{3M} \ell}{N \gamma (b_{3M} \ell + \mu) \mu^2}}, \quad (22)$$

where $\mathcal{R}_0^H = \frac{ab_2}{\gamma}$ and $\mathcal{R}_0^M = \frac{ab_1 \Lambda}{N \mu} \cdot \frac{b_{3M} \ell}{(b_{3M} \ell + \mu) \mu}$.

5.3. Understanding relations

Having the necessary epidemiological and environmental datasets ready for use, the next step taken was to verify the correlation between the number of cases and forest areas impacted by deforestation and fires, in addition to the correlation with temperature and precipitation as well.

Using the Pandas library on a Python environment, the correlation between monthly number of cases caused by *P. vivax* in Manaus and deforested area was found to be approximately -0.11, and 0.137 between cases and burnt area. In relation to climatic data, the correlation with temperature was found to be around 0.404, and with precipitation to be -0.365. Both correlation tables were taken in relation to monthly data.

On the other hand, taking the correlation between the incidence of cases in relation to the seasons, deforested and burnt areas, that was found to be 0.722, 0.314 and 0.818, respectively. Looking for the correlation in relation to seasons, temperature and precipitation, that was found to be 0.687, 0.839 and -0.883, respectively.

6. Computational Implementation

6.1. Numerical Solver

All model simulations were performed using Python version 3.12.7 (conda-forge distribution). Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

The systems of ODEs (5a-g) were solved numerically using the LSODA solver as implemented in the `solve_ivp` function from SciPy version 1.17.1. All computations were performed on a laptop running **Windows 11** (64-bit) equipped with a **12th Gen Intel Core i5-1235U** processor (10 cores, 12 threads, 1.30 GHz base frequency) and **16 GB RAM**.

The source code and data required to reproduce all figures are available at https://github.com/RaphaLevy/Study_on_malaria_dynamics_v2.

6.2. Calibration and Parameter Estimation

6.2.1. Objective function

Model parameters were estimated by minimizing the discrepancy between the simulated infected-human trajectory $I_H(t)$ and the observed active-case counts reported through SIVEP-Malaria, restricted to rural Manaus (scaled by the 41.54% rural proportion). The objective function is a weighted mean squared error (wMSE):

$$\text{wMSE}(\theta) = \frac{\sum_{i=1}^n w_i (I_H(t_i; \theta) - y_i)^2}{\sum_{i=1}^n w_i}, \quad (23)$$

where y_i denotes the observed active-case count at date t_i , $I_H(t_i; \theta)$ is the model prediction obtained by linear interpolation of the simulated trajectory onto the observation dates, and θ is the vector of calibrated parameters.

The weights w_i follow an exponential decay that prioritizes early-time dynamics within each year:

$$w_i = w_0 \exp(-\lambda d_i), \quad (24)$$

where d_i is the number of days from the start of the year for observation i , $w_0 = 5.0$ is the initial weight, and $\lambda = 0.0015 \text{ day}^{-1}$ is the decay rate. This weighting scheme reflects the greater sensitivity of the early-season trajectory to initial conditions and parameter values, guiding the optimizer toward solutions that correctly capture the onset of the rainy-season transmission peak.

To prevent the optimizer from collapsing the infected-mosquito population to near-zero levels—which would sever the force-of-infection coupling and render the human transmission parameters structurally unidentifiable—an extinction penalty was added to the objective:

$$J(\theta) = \text{wMSE}(\theta) + \mathcal{P}(I_M^{\min}(\theta)), \quad (25)$$

where $I_M^{\min}(\theta) = \min_t I_M(t; \theta)$ is the annual minimum of the infected-mosquito compartment, and

$$\mathcal{P}(I_M^{\min}) = \begin{cases} 0 & \text{if } I_M^{\min} \geq I_M^*, \\ B \left(1 + \frac{I_M^* - I_M^{\min}}{I_M^*} \right) & \text{if } I_M^{\min} < I_M^*, \end{cases} \quad (26)$$

with $I_M^* = 10$ (viability threshold) and $B = 10^8$ (penalty base scale). The penalty scales linearly with the depth of extinction, ensuring that partially-extinct trajectories are ranked proportionally rather than receiving a uniform penalty that would flatten the objective landscape.

6.2.2. Parameter grouping and bounds

The nine calibrated parameters were grouped into four biological “topics” for sequential fitting:

1. **Humidity response** (H_0, k, φ): the half-saturation threshold, sigmoid steepness, and baseline survival floor of the humidity-dependent survival function $p(H)$;
2. **Human transmission** (τ_H, γ, ω): the intrinsic incubation period, recovery rate, and immunity waning rate;
3. **Force of infection** (b_1, b_2): the human-to-mosquito and mosquito-to-human transmission probabilities;
4. **Carrying capacity** (M'): the baseline mosquito population scale that determines the environmental carrying capacity $K(t)$.

The parameter bounds are given in Table 1. The lower bound on $H_0 = 20\%$ prevents the humidity response from collapsing (which would drive $\mu \rightarrow \infty$ and extinguish the vector population). The bound on $b_2 \geq 0.01$ ensures a minimum level of mosquito-to-human transmission, while $\tau_H \in [10, 20]$ days constrains the human incubation period to the biologically plausible range for *P. vivax* [37].

Table 1

Parameter bounds for sequential topic calibration.

Parameter	Lower bound	Upper bound
H_0 (%)	20	75
k ($\%^{-1}$)	0.01	1.0
φ	0.01	0.5
τ_H (days)	10	20
γ (days^{-1})	1/150	1.0
ω (days^{-1})	1/300	0.1
b_1	0.001	0.5
b_2	0.01	0.5
M'	50,000	$500 \times N_y$

N_y denotes the estimated rural population of Manaus for year y .

6.2.3. Year-by-year sequential calibration

Parameters were calibrated sequentially for each year from 2017 to 2023 using the `lmfit` implementation of differential evolution (DE) [38]. The DE settings were: population multiplier `popsz` = 6, maximum function evaluations `max_nfev` = 800, convergence tolerance `tol` = 0.01, and random seed fixed at 42 for reproducibility.

Within each year, the four topics were fitted in the fixed order:

$$\underbrace{(H_0, k, \varphi)}_{\text{humidity}} \rightarrow \underbrace{(\tau_H, \gamma, \omega)}_{\text{human}} \rightarrow \underbrace{(b_1, b_2)}_{\text{FoI}} \rightarrow \underbrace{M'}_{\text{capacity}}. \quad (27)$$

At each step, only the current topic’s parameters were free; all previously fitted parameters were held fixed at their most recent estimates. This sequential strategy reduces the dimensionality of each optimization sub-problem while respecting the biological ordering: the humidity response determines mosquito mortality, which constrains the infection cycle, which in turn determines the force of infection and carrying capacity.

After the sequential pass, a joint refinement was performed using the Nelder-Mead simplex method [39], with all nine parameters free simultaneously, seeded from the sequential solution. A seed-evaluation guard was applied: after each DE run and after each refinement stage, the starting point was explicitly evaluated, and the optimizer’s output was accepted only if it improved upon the seed. This prevents the stochastic DE from returning a solution worse than its initialization.

6.2.4. Year chaining and initial conditions

The model was calibrated sequentially across years, with the end-state vector of each year serving as the initial state for the next:

$$\mathbf{z}_0^{(y+1)} = \mathbf{z}_{\text{end}}^{(y)}, \quad (28)$$

where $\mathbf{z} = (S_H, E_H, I_H, R_H, S_M, E_M, I_M)$ is the full state vector. The human infected compartment was always

reset to the first observed active-case count for the new year:

$$I_{H,0}^{(y+1)} = \text{round}(y_{\text{first}}^{\text{obs}}), \quad (29)$$

while all other compartments were carried forward without modification. Negative end-state values were clamped to zero before carrying.

For the first year (2017), initial-condition fractions were explicitly fitted using Nelder-Mead with 300 function evaluations:

$$S_{H,0} = \alpha_S \cdot N, \quad E_{H,0} = \alpha_E \cdot N, \quad I_{M,0} = \alpha_I \cdot M_0, \quad (30)$$

where $\alpha_S \in [0.3, 0.9]$, $\alpha_E \in [0.001, 0.30]$, and $\alpha_I \in [0.0005, 0.15]$, with the remaining compartments determined by conservation: $R_{H,0} = N - S_{H,0} - E_{H,0} - I_{H,0}$, $M_0 = 10N$, and $S_{M,0} = M_0 - E_{M,0} - I_{M,0}$.

6.2.5. Weekly diagnostic

To evaluate where the climate-driven formulation deviates from what the data demands, a diagnostic calibration was performed in which the effective transmission coefficients were freed as per-week parameters instead of being derived from the biting rate and fixed b_1, b_2 :

$$\lambda_H(t) = \beta_{h,w} \frac{I_M(t)}{N}, \quad \lambda_M(t) = \beta_{m,w} \frac{I_H(t)}{N}, \quad (31)$$

where $w = \lfloor d/7 \rfloor$ is the week index and d is the day number within the year. This yields $2 \times n_{\text{weeks}}$ free parameters per year (typically 106 for a 365-day year), with bounds $\beta_{h,w}, \beta_{m,w} \in [10^{-9}, 3.0]$.

All other model parameters were fixed at the values obtained from the yearly topic calibration. The initial state was reconstructed from the previous year's end-state, with I_H reset to observed data. The optimizer used was `lmfit.minimize` with the trust-region reflective method (`least_squares`), with `max_nfev = 1200` and `ftol = 10-8`.

The residual vector for the trust-region solver was constructed as:

$$r_i = \sqrt{w_i} (I_H(t_i) - y_i) + \sqrt{\mathcal{P}(I_M^{\min})}, \quad (32)$$

where $\sqrt{\mathcal{P}}$ is injected as a constant offset to preserve the residual vector length while providing a gradient that pushes I_M upward when the mosquito population approaches the viability threshold.

The weekly beta fit serves as a diagnostic baseline: it reveals the best achievable fit when the transmission coefficients are unconstrained by the climate formulation. Persistent deviations between the weekly-optimal $\beta_{h,w}$ and the climate-driven $a(T) \cdot b_2$ indicate periods where the environmental model fails to capture the true transmission dynamics.

7. Results

7.1. Reproduction of the Original Parham & Michael Model

As a baseline for the development of the model, the original SIR–SEI proposed by Parham and Michael (2010) was implemented using all parameter values reported in the original study. Since population sizes and initial epidemiological conditions were not explicitly provided by the authors, these quantities were estimated from the available rural Manaus data and used as initial conditions.

Figures 1A and 2A show the resulting dynamics of the human and mosquito populations, respectively. While the epidemiological compartments for the human population initially exhibited plausible behavior, although not what would be expected for a recurring disease such as malaria, with the infected population gradually decreased, the mosquito population rapidly declined to extinction. This behavior indicates that mosquito recruitment was insufficient to replace mortality.

Since transmission depends on the persistence of infectious mosquitoes, the collapse of the vector population resulted in the elimination of infection in both host and vector populations. Therefore, although the delay formulation preserved positivity, it failed to reproduce the endemic behavior expected for malaria transmission in the Amazon region.

8. Discussion

The population estimates provided by the Trajetórias Project [6] for inter-census periods are derived using fixed rural–urban proportions based on the 2010 census. While this approach ensures internal consistency, it does not account for the significant demographic changes observed in Manaus over time, particularly those associated with rapid urbanization.

To address this limitation, the population proportions used in this study were recalculated (see Section 3.3) using interpolation and extrapolation based on available census data. This adjustment provides a more realistic representation of temporal changes in the rural population, which is especially relevant given the disproportionate contribution of rural areas to malaria incidence.

These refinements improve the consistency between demographic inputs and epidemiological dynamics in the model, reducing potential biases in the estimation of transmission intensity and enhancing the reliability of the results.

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A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$p(T)$	Daily mosquito survival rate dependent on temperature	$e^{(-1/(AT^2+BT+C))}$
$p(H)$	Daily mosquito survival rate dependent on humidity ²	$\frac{\varphi e^{-k(H-H_0)+1}}{e^{-k(H-H_0)+1}+1}$
$p(T, H)$	Daily mosquito survival rate	$p(T)p(H)$
$p_L(R)$	Probability of larval survival dependent on rainfall ³	$(4p_{ML}/R_L^2)R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1T+c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R)p_L(T)$
$\mu(T, H)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T, H))$
$a(T)$	Biting rate (days ⁻¹)	$(T - T')/D_1$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD/(T - T_{\min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1/(c_1T + c_2)$
$\ell(\tau_M)(T, H)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T, H)^{\tau_M(T)}$
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$M(t)$	Total population of mosquitos	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + E_H(t) + I_H(t) + R_H(t)$

² $p(H)$ was added in order to take into consideration the deleterious effects of humidity. Unlike for T and R , which use the daily average value of temperature and precipitation, respectively, H uses the daily minimal value of humidity.

³ $p_E(R)$ and $p_P(R)$ have an analogous expression to $p_L(R)$, substituting p_{ML} for p_{ME} and p_{MP} respectively.

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection	Dimensionless
γ	1/Average duration of infectiousness in humans	days ⁻¹
ω	1/Average duration of protective immunity after infection	days ⁻¹
B_E	Number of eggs laid per adult per oviposition	Dimensionless
p_{ME}	Maximum probability of egg survival	Dimensionless
p_{ML}	Maximum probability of larval survival	Dimensionless
p_{MP}	Maximum probability of pupal survival	Dimensionless
τ_E	Duration of the egg development phase	days
τ_P	Duration of the pupal development phase	days
τ_H	Latent period of infection within humans	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 ⁴	mm
$T_{\min}$	Minimum temperature, below which there is no development of the parasite: 14.5	°C
$T_{\max}$	Maximum temperature, above which there is no development of the parasite: 29.8 [9]	°C
T_{opt}	Optimal temperature for development of the parasite: 23.5 [9]	°C
$T_{\min A.}/T'$	Minimum temperature, below which there is no development of the vector: 16 ⁵ [42, 44]	°C
$T_{\max A.}$	Maximum temperature, above which there is no development of the vector: 34 [42, 44, 45]	°C
$T_{\text{op } A.}$	Optimal temperature for transmission: 34 [42]	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [9, 10]	°C days
A	Empirical sensitivity parameter	(°C ² days) ⁻¹
B	Empirical sensitivity parameter	(°C days) ⁻¹
C	Empirical sensitivity parameter	days ⁻¹
D_1	Temperature scaling factor of the biting rate	°C days
c_1	Empirical sensitivity parameter	(°C days) ⁻¹
c_2	Empirical sensitivity parameter	days ⁻¹
b_{3M}	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
b_{3H}	Infection rate in exposed humans ($1/\tau_H$)	days ⁻¹
φ	Baseline survival probability at extreme low humidity	Dimensionless
H_0	Critical humidity threshold for 50% survival transition	%
k	Steepness of the humidity-survival sigmoid response	% ⁻¹

⁴ The reference rainfall level R_L for the Amazon was approximated based on reported associations between monthly precipitation and malaria incidence [8]. In particular, precipitation levels in the range of 917 to 1019 mm/month were associated with a decrease in malaria cases. The midpoint of this interval (968 mm/month) was used as a representative value and converted to a daily rate by dividing by 30 days, yielding $R_L \approx 32.3$ mm/day. This value is therefore interpreted as the threshold scale for high rainfall conditions in the model.

⁵ In Parham and Michael [7], this parameter is denoted by T_1 . Here, we adopt a different notation to avoid confusion with the mean temperature in the absence of seasonality, also denoted by T_1 in the original model. Additionally, instead of defining an independent lower thermal threshold for the biting rate, we assume that biting activity becomes negligible below the minimum temperature required for vector development.

A.2. Figures

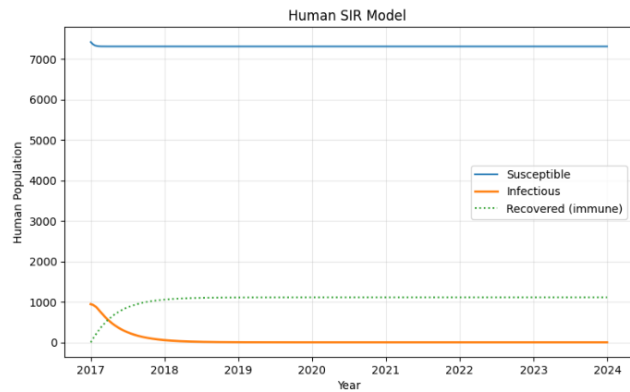


Figure 1A: DDE simulation of the baseline SIR model using the original parameterization from Parham & Michael (2010) [7]. The dynamics show a monotonic decline in the number of infected individuals, indicating the absence of sustained transmission.

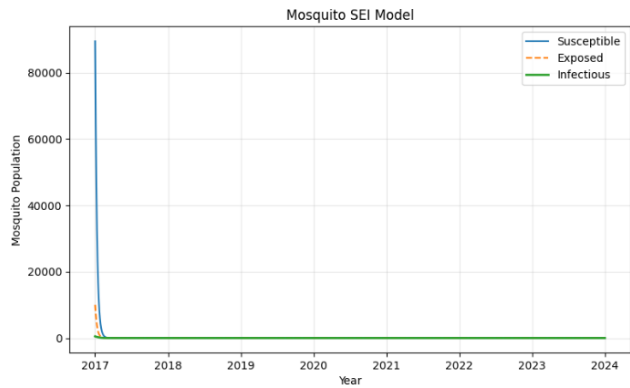


Figure 2A: DDE simulation of the baseline SEI mosquito model using the original parameterization from [7]. The dynamics show that the entire population tended to extinction, an unexpected behavior for a region where malaria is highly endemic.

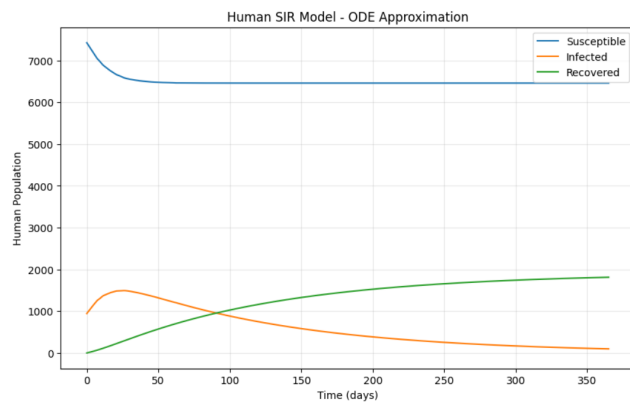


Figure 3A: ODE simulation of the baseline SIR model using the formulation from 2.2 and parameters from [7]. A slight increase in the infectious population can be noticed, although it still tends to extinction over the course of the first year of transmission.

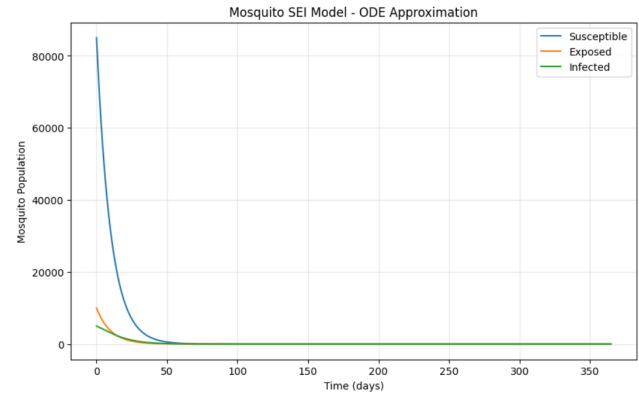


Figure 4A: ODE simulation of the baseline SEI mosquito model using the formulation from 2.2 and parameters from [7]. It can be seen that the mosquito population still tends to extinction.